

ClimateScope

Milestone 1: Data Preprocessing and Initial Analysis Report

1. Introduction

The purpose of this milestone is to ensure the raw weather dataset is transformed into a structured, clean, and analysis-ready format. This step is crucial for building reliable models, maintaining data integrity, and supporting accurate exploratory data analysis.

2. Dataset Overview

The dataset consists of raw weather-related variables collected over multiple timestamps. Initial inspection indicated the presence of missing values, inconsistencies, and numerical features requiring normalization.

3. Tasks Performed

3.1 Identification of Missing Values and Inconsistencies

- Performed an audit of the dataset to check for NaNs and incomplete entries.
- Identified columns with significant missing values and potential outliers.
- Verified consistency of units and data ranges (e.g., temperature values, humidity %, wind speed).

3.2 Handling Missing Data

- Cleaned the dataset by handling NaN values appropriately.
- Applied imputation strategies such as:
 - Mean/median replacement for numerical features.
 - Forward-fill/backward-fill where sequential continuity was necessary.

This preprocessing ensures the dataset does not contain gaps that could impact downstream analysis.

3.3 Normalization and Aggregation of Numerical Features

- Applied **Min-Max Scaling** to normalize the numerical variables.
- Transformed all continuous features into a uniform range (0 to 1).
- Ensured that features with different scales (e.g., temperature vs. wind speed) now contribute proportionally in further analysis and modeling.

4. Outcome

After performing the above tasks:

- The dataset is now fully cleaned and free from missing or invalid values.
- All numerical fields are normalized and standardized for analysis.
- The processed dataset is ready for Exploratory Data Analysis (EDA) and model development in subsequent milestones.

5. Conclusion

The preprocessing activities completed in this milestone significantly enhance data quality and usability. With clean, normalized, and consistent data, reliable insights can now be derived in the next phases of the project.